WEEK 1

DATA CLEANING & FEATURE ENGINEERING REPORT

Week 1: Data Cleaning & Feature Engineering Report

Team: 1011 – AI Group 10

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Final Dataset: *production_ready_dataset_v2*.xlsx

Total Records: 8,558

1. Introduction

Purpose

This report summarizes all Week 1 work completed as part of the AI Data Analyst Internship. The objective was to transform Excelerate's raw learner engagement dataset into a clean, reliable, and analysis-ready format through systematic data preprocessing, validation, and feature engineering. These activities form the foundation for Week 2 Exploratory Data Analysis (EDA) and subsequent modeling tasks.

Dataset Overview

The original dataset included learner demographics, opportunity lifecycle data, timestamps, status codes, and engagement details. The data originated from multiple sources and contained inconsistencies such as mixed date formats, corrupt timestamps, missing values, and duplicated category labels. The cleaned dataset consists of **8,558 records** and retains 100% of the original rows.

2. Data Cleaning Process

Week 1 followed a structured pipeline consisting of three major phases: normalization, imputation, and quality validation.

Phase 1 — Data Normalization

1. Column Name Normalization

- Converted all column names to standardized snake_case.
- Removed characters such as spaces, slashes, and hyphens.
- Achieved consistent naming across 16 original columns.

2. Date Parsing and Recovery

- Parsed 6 major date-time columns containing inconsistent formats (MM/DD/YYYY, DD/MM/YYYY, YYYY-MM-DD).
- Removed **735 corrupt time stamps** (e.g., 708:21:29 with invalid hour value).
- Applied a multi-pass parsing strategy using dayfirst=True/False and explicit format matching.
- Converted all valid timestamps to datetime64[ns].
- **Parsing success rate achieved: 99.2%.**

3. Text Standardization

- Normalized text casing using title case across fields such as first_name, country, and status_description.
- Removed extra whitespace and trailing punctuation.
- Ensured uniform text encoding.

4. Categorical Value Normalization

- **Gender:** Re-mapped variations (M, m, Male, male, etc.) into:
  Male, Female, Other, Prefer Not To Say
  Missing values → "Unknown".
- **Status Descriptions:** Standardized labels including: Started, Team Allocated, Rejected, Withdrawn.
- **Institution Names:** Corrected inconsistent spellings (e.g., "Saint Louis University").
- **Country:** Standardized naming conventions across all rows.

Phase 2 — Missing Value Handling

5. Missing Value Analysis

- Identified **10,754 missing values** (~5.71% of all cells).
- Mapped missingness patterns by column and opportunity group.

6. Hybrid Imputation Strategy

A combination of logical replacement, forward/backward propagation, and recalculation was used:

- **Group-Based Start/End Date Filling**
  For each opportunity_id, missing start/end dates were forward/backward filled.
  - Restored **3,164 missing cells:**
    - 1,902 opportunity_start_date
    - 1,262 opportunity_end_date
- **Recalculation of Derived Fields:**
  After imputation, all date-based features were recalculated, including: engagement_lag_days, opportunity_duration_days, days_before_start.
- **Sparse Text Field Imputation:**
  Filled infrequent missing text values with "Unknown".
- **Final Improvement:**
  Completeness improved by **29.5%**, reaching **94.29%** before feature engineering.

Phase 3 : Data Quality Validation

7. Duplicate Detection

- Performed a full scan for exact duplicate rows.
- **Result:** No duplicates found → 100% unique records.

8. Chronology Validation

- Identified **735 cases** where apply_date < signup_datetime (impossible).
  → Marked with flag_engagement_inversion = 1.
- Verified opportunity_start_date ≤ opportunity_end_date (no violations).
- Identified one unrealistic age >120 years → corrected to NaN.

9. Negative Value Inspection

- Found **735 negative engagement_lag_days**, caused by chronology inversions.
  → Set to NaN and flagged.
- Valid negative values in days_before_start (late applications) were retained.

10. Outlier Analysis

- Detected **394 extreme lead/lag cases** in days_before_start (± >365 days).
- Marked these as flag_days_before_start_extreme = 1.

11. Data Type Standardization

- Ensured all date fields → datetime64[ns].
- Integer fields → Int64.
- Floats → float64.
- Categorical text → object.
- **Result:** Achieved 100% type consistency.

3. Feature Engineering

A total of **11 engineered features** were added to enhance analytical and modeling readiness.

Core Engineered Features

1. age_years

- **Formula:** floor((signup_date – DOB)/365.25)
- **Range:** 12–57
- **Purpose:** Demographic segmentation
- **Missing:** 3.5%

2. signup_month

- Extracted month from signup timestamp.
- Useful for seasonal & monthly engagement trends.

3. signup_year

- Extracted year from signup timestamp.
- Distribution: 2023 (62%), 2024 (38%).

4. engagement_lag_days

- **Formula:** (apply_date – signup_datetime).days
- **Range:** 0–695
- **Missing:** 14.8% (includes invalid chronology cases)
- **Used for:** Measuring application speed.

5. opportunity_duration_days

- **Formula:** end_date – start_date
- Range: 0–706
- Median duration: 300 days
- **Purpose:** Understanding opportunity length.

6. days_before_start

- **Formula:** start_date – apply_date
- Positive → applied early
- Negative → applied after start

- Range: –761 to +445
- **Purpose:** Application timing analysis.

Derived Features & Quality Flags

7. engagement_lag_bucket

Categorical segmentation of lag days:
0, 1–7, 8–30, 31–90, 90+, Unknown.

8. applied_after_start

Binary field (1 = applied after start).

9. log_opportunity_duration

Log-scaled duration to reduce skew.

10. flag_engagement_inversion

Flags invalid chronology cases (735 records).

11. flag_days_before_start_extreme

Flags extreme timing outliers (>1 year).

12. engagement_lag_days_fixed

Cleaned version of engagement_lag_days after corrections.

4. Specific Issues Encountered & Resolved

Issue 1 — Chronology Inversions

- **735 records** had apply_date before signup_date.
- Set lag to NaN and flagged with flag_engagement_inversion.
- Full traceability logged.

Issue 2 — Corrupt Timestamps

- Timestamps like "708:21:29" corrupted parsing.
- Resolved using regex removal + multi-pass parsing.
- All 735 corrupt timestamps corrected.

Issue 3 — Out-of-Range Age

- One DOB produced age >120 years.

- Corrected to NaN after validation.

Issue 4 — Missing Opportunity Dates

- Over 3,100 date fields missing.
- Restored using group-based ffill/bfill.

Issue 5 — Deprecated Methods

- Updated all outdated pandas fillna(method=...) to .ffill()/.bfill().

Issue 6 — Index Alignment Errors

- Sorting prior to groupby caused index mismatch.
- Removed sorting and used correct transform alignment.

Issue 7 — Extreme Timing Values

- 394 cases applied extremely early/late.
- Flagged for modeling awareness.

5. Data Validation Summary

Final Dataset Quality

- **Completeness:** 93.63% overall
- **Type Consistency:** 100%
- **Duplicate Rows:** 0
- **Invalid Values:** Corrected or flagged
- **Record Retention:** 100% (all 8,558 rows preserved)

The dataset is now stable, coherent, and suitable for visualization, predictive modeling, and advanced analysis.

6. Conclusion & Week 2 Preview

Week 1 successfully delivered a **cleaned, validated, and feature-rich dataset** supported by a full audit trail. All major structural issues were resolved, enabling a strong foundation for Week 2.

Next Steps for Week 2

- Exploratory Data Analysis (EDA)
- Trend and cohort visualizations
- Engagement funnel analysis
- Correlation and feature importance exploration

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7. Appendix

- **Final dataset:** production_ready_dataset_v2.xlsx
- **GitHub Repository :** All scripts, cleaning functions, audit logs, validation outputs, and documentation used throughout the Week 1 pipeline are maintained in the project's GitHub repository:

  🔗 **Repository Link:** **https://github.com/Prajwal-031/AI-Data-Analyst-Internship-End-to-End-Data-Cleaning-ETL-Pipeline**

  This repository provides a complete, production-style data engineering and analytics workflow developed during the AI Data Analyst Internship (Weeks 1–4). It includes the full end-to-end pipeline for ingesting messy source files, performing systematic cleaning and normalization, executing feature engineering, validating outputs, and generating final analysis-ready datasets. The repository also contains all supporting resources, including the README, scripts, audit trails, intermediate outputs, and documentation for reproducibility.

Tools & Technologies Used — List Format

Programming & Data Processing

- **Python 3.10+**
- **Pandas**
- **NumPy**
- **Regex (re module)**

Development & Execution Environments

- **Visual Studio Code (VS Code)**

Data Inspection & File Handling

- **Microsoft Excel**
- **CSV & Excel (.xlsx) file formats**

Version Control & Collaboration

- **GitHub**

Validation & Testing

- **Custom Python Unit Tests**
- **Audit Logs (CSV-based):**
    - **fixes_audit.csv**
    - **engagement_lag_inversions_audit.csv**

Documentation & Reporting

- **Microsoft Word**
- **PDF Export Tools**